R20

Code No.: 20CSC21

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (CSE) V Sem (Main & Backlog) Examination Dec 2023 – Jan 2024

Data Communication and Computer Networks

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		What is the number of cable links required for mesh topology? If 5 nodes are connected in a network using mesh topology the number of cables required to connect them?	(3)	1	1
2		What is the advantage of selective repeat ARQ protocol over Go-Back-N ARQ protocol?	(3)	2	2
3		What is Count-to-Infinity problem in Distance Vector Routing. List two approaches to overcome the same.	(3)	3	2
4		What is four way handshaking? In which protocol it will be used.	(3)	5	3
5		Differentiate between IMAP and POP. Discuss with one real-time example.	(3)	6	3

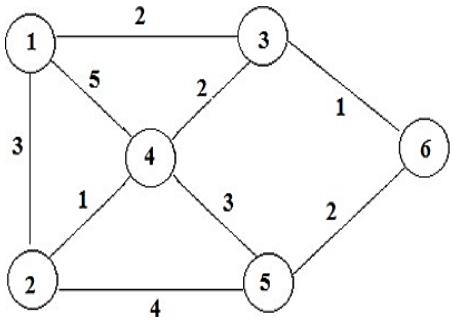
Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6		Draw the neat diagram of complete OSI reference model layered architecture and discuss about the responsibilities of each layer in brief.	(9)	1	1
		(OR)			
7	(a)	Discuss about the guided media and explain with real – time applications for the below: i) Twisted pair ii) Coaxial cable and iii) Fiber optical cable	(5)	1	3
	(b)	Describe bus topology with suitable example and give their advantages and disadvantages.	(4)	1	1
8	(a)	Differentiate the flow control and error control. Discuss the importance of error control in transmission.	(3)	2	2
	(b)	Let assume data frame has eight bits, calculate the number of padding bits to detect and correct the data according to hamming code and generate the transmitted frame. The forth bit in dataword is inverted during transmission so find the error position and correct the error and justify the error is single bit error.	(6)	2	2
		(OR)			
9	(a)	What is CSMA/CD? How does it work?	(5)	2	2
	(b)	Draw the HDLC protocol frame format and analyze 8 – bit control field in I, S and U frames.	(4)	2	2

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10	(a)	What is the format, bit representation, and address space for the following IP addressing. i) IPv4 ii) IPv6	(4)	3	3
	(b)	Discuss the role of Internet Control Message Protocol? List different Internet Control Message Protocol message types.	(5)	3	3
		(OR)			
11	(a)	Define optimality principle? Illustrate the Dijkstra's shortest path routing algorithm for the given graph and find the shortest path from node 1 to node 6.	(6)	4	3



	(b)	What is flooding? Explain it with an example also mention the advantages and disadvantages of flooding?	(3)	4	3
12	(a)	Compare the TCP header and the UDP header.	(4)	5	2
	(b)	What is three-way handshake mechanism in TCP/IP and how connection establish and connection release taken place in it.	(5)	5	2
		(OR)			
13	(a)	Explain the connection establishment in SCTP with an example.	(6)	5	2
	(b)	What is association in SCTP? Discuss about the advantages of SCTP over TCP and UDP.	(3)	5	2
14	(a)	In the Simple Network Management Protocol mention the working principle and related issues of the SNMP?	(6)	6	2
	(b)	Differentiate the FTP and SFTP	(3)	6	2
		(OR)			
15	(a)	What is World Wide Web? Brief on the architectural overview of the world wide web?	(5)	6	2
	(b)	Give the Architectural lay out of the Electronic mail (E-mail)? In email what is the functionality of the User Agent (UA) and Message Transfer Agent (MTA)?	(4)	6	2

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